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Day-12

Python Arrays

- Python Arrays
- An array is defined as a collection of items that are stored at contiguous memory locations. It is a container which can hold a fixed number of items, and these items should be of the same type. An array is popular in most programming languages like C/C++, JavaScript, etc.

- Array is an idea of storing multiple items of the same type together and it makes easier to calculate the position of each element by simply adding an offset to the base value. A combination of the arrays could save a lot of time by reducing the overall size of the code. It is used to store multiple values in single variable. If you have a list of items that are stored in their corresponding variables like this:

- car1 = "Lamborghini"

- car2 = "Bugatti"
- car3 = "Koenigsegg"
- If you want to loop through cars and find a specific one, you can use the array.
- The array can be handled in Python by a module named array. It is useful when we have to manipulate only specific data values. Following are the terms to understand the concept of an array:
- Element - Each item stored in an array is called an **element**.
- **Index -** The location of an element in an array has a numerical index, which is used to identify the position of the element.

- Array Representation
- An array can be declared in various ways and different languages. The important points that should be considered are as follows:

- Index starts with 0.
- We can access each element via its index.
- **The length of the array defines the capacity to store the elements**.

Array operations

- Some of the basic operations supported by an array are as follows:

- **Traverse** - It prints all the elements one by one.
- **Insertion - It adds an element at the given index.**
- **Deletion** - It deletes an element at the given index.
- Search - It searches an element using the given index or by the value.
- Update - It updates an element at the given index.
- The Array can be created in Python by importing the array module to the python program.

- from array import *
- arrayName = array(typecode, [initializers])

Accessing array elements

- We can access the array elements using the respective indices of those elements.

- **import array as arr**
- **a = arr.array('i', [2, 4, 6, 8])**
- **print("First element:", a[0])**
- **print("Second element:", a[1])**
- **print("Second last element:", a[-1])**
- **Output:**

- First element: 2
- Second element: 4
- Second last element: 8.
- **Explanation:** In the above example, we have imported an array, defined a variable named as "a" that holds the elements of an array and print the elements by accessing elements through indices of an array.

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- **How to change or add elements**
- Arrays are mutable, and their elements can be changed in a similar way like lists.

- import array as arr
- numbers = arr.array('i', [1, 2, 3, 5, 7, 10])
-
- # changing first element
- numbers[0] = 0
- print(numbers) # Output: array('i', [0, 2, 3, 5, 7, 10])
-
- # changing 3rd to 5th element
- numbers[2:5] = arr.array('i', [4, 6, 8])
- print(numbers) # Output: array('i', [0, 2, 4, 6, 8, 10])

- Output:

- array('i', [0, 2, 3, 5, 7, 10])
- array('i' ,[0, 2, 4, 6, 8, 10])
- Explanation: In the above example, we have imported an array and defined a variable named as "numbers" which holds the value of an array. If we want to change or add the elements in an array, we can do it by defining the particular index of an array where you want to change or add the elements.

Why to use arrays in Python?

- A combination of arrays saves a lot of time. The array can reduce the overall size of the code.

- How to delete elements from an array?
- The elements can be deleted from an array using Python's del statement. If we want to delete any value from the array, we can do that by using the indices of a particular element.

- import array as arr
- number = arr.array('i', [1, 2, 3, 3, 4])
- del number[2]   # removing third element
- print(number)   # Output: array('i', [1, 2, 3, 4])
- Output:

- array('i', [10, 20, 40, 60])
- Explanation: In the above example, we have imported an array and defined a variable named as "number" which stores the values of an array. Here, by using del statement, we are removing the third element [3] of the given array.

Finding the length of an array

- The length of an array is defined as the number of elements present in an array. It returns an integer value that is equal to the total number of the elements present in that array.
- Syntax
- len(array_name)
- **Array Concatenation**
- **We can easily concatenate any two arrays using the + symbol.**
- **Example**

- a=arr.array('d',[1.1 , 2.1 ,3.1,2.6,7.8])
- b=arr.array('d',[3.7,8.6])
- c=arr.array('d')
- c=a+b
- print("Array c = ",c)
- Output:
- Array c= array('d', [1.1, 2.1, 3.1, 2.6, 7.8, 3.7, 8.6])
- Explanation
- In the above example, we have defined variables named as "a, b, c" that hold the values of an array.

- Example

- **import array as arr**
- **x = arr.array('i', [4, 7, 19, 22])**
- **print("First element:", x[0])**
- **print("Second element:", x[1])**
- **print("Second last element:", x[-1])**
- **Output:**

- **First element: 4**
- **Second element: 7**
- **Second last element: 22**
- **Explanation: In the above e**xample, first, we have imported an array and defined a variable named as "x" which holds the value of an array and then, we have printed the elements using the indices of an array.

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Day Tasks

- Exercise 1:This code employs slicing to extract elements or subarrays from an array. Perform and verify the output
- import array as arr
- l = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

- a = arr.array('i', l)
- print("Initial Array: ")
- for i in (a):
-     print(i, end=" ")
- Sliced_array = a[3:8]
- print("\nSlicing elements in a range 3-8: ")
- print(Sliced_array)
- Sliced_array = a[5:]
- print("\nElements sliced from 5th "
-       "element till the end: ")
- print(Sliced_array)
- Sliced_array = a[:]
- print("\nPrinting all elements using slice operation: ")
- print(Sliced_array)

Excercise-2

- Perform the code demonstrates how to create array in Python, print its elements, and find the indices of specific elements.
- import array
- arr = array.array('i', [1, 2, 3, 1, 2, 5])
- print("The new created array is : ", end="")
- for i in range(0, 6):
-     print(arr[i], end=" ")

- print("\r")
- print("The index of 1st occurrence of 2 is : ", end="")
- print(arr.index(2))
- print("The index of 1st occurrence of 1 is : ", end="")
- print(arr.index(1))

Excercise-3

- import array
- arr = array.array('i', [1, 2, 3, 1, 2, 5])
- print("Array before updation : ", end="")
- for i in range(0, 6):
-     print(arr[i], end=" ")

- print("\r")
- arr[2] = 6
- print("Array after updation : ", end="")
- for i in range(0, 6):
-     print(arr[i], end=" ")
- print()
- arr[4] = 8
- print("Array after updation : ", end="")
- for i in range(0, 6):
-     print(arr[i], end=" ")

This PPT contents are

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